Предел функции

Вариант 1

ЧБ 17
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№1

$$\lim_{x\to 3}\frac{x^2-9}{x^2-5x+6}=\left[\frac{0}{0}\right]=\lim_{x\to 3}\frac{(x-3)(x+3)}{(x-3)(x-2)}=$$

$$=\lim_{x\to 3}\frac{x+3}{x-2}=6$$

№2

$$\lim_{x\to 2}\frac{x^3-3x^2-4x+12}{x^3-2x^2-x+2}=\left[\frac{0}{0}\right]=\lim_{x\to 2}\frac{x^2(x-3)-4(x-3)}{x^2(x-2)-(x-2)}=$$

$$=\lim_{x\to 2}\frac{(x-3)(x^2-4)}{(x-2)(x^2-1)}=\lim_{x\to 2}\frac{(x-3)(x-2)(x+2)}{(x-2)(x-1)(x+1)}=$$

$$=\lim_{x\to 2}\frac{(x-3)(x+2)}{(x-1)(x+1)}=\lim_{x\to 2}\frac{-1\cdot 4}{1\cdot 3}=-\frac{4}{3}$$

№3

$$\lim_{x\to 4}\frac{\sqrt{x+5}-3}{x-4}=\left[\frac{0}{0}\right]=\lim_{x\to 4}\frac{(\sqrt{x+5}-3)(\sqrt{x+5}+3)}{(x-4)(\sqrt{x+5}+3)}=$$

$$=\lim_{x\to 4}\frac{(x+5)-9}{(x-4)(\sqrt{x+5}+3)}=\lim_{x\to 4}\frac{x-4}{(x-4)(\sqrt{x+5}+3)}=\lim_{x\to 4}\frac{1}{\sqrt{9}+3}=\frac{1}{6}$$

№4

$$\lim_{x\to+\infty}\frac{5x^3-2x+1}{-2x^3+7x^2-4}=\left[\frac{\infty}{\infty}\right]=\lim_{x\to+\infty}\frac{x^3\left(5-\frac{2}{x^2}+\frac{1}{x^3}\right)}{x^3\left(-2+\frac{7}{x}-\frac{4}{x^3}\right)}=$$

($\frac{2}{x^2}\to 0$, $\frac{1}{x^3}\to 0$, $\frac{7}{x}\to 0$, $\frac{4}{x^3}\to 0$)

$$=\lim_{x\to\infty}\frac{5}{-2}=-2{,}5$$

N5

$$\lim_{x\to\infty}\left(\sqrt{x^2+5x+1}-\sqrt{x^2-x+2}\right)=[\infty-\infty]=$$

$$=\lim_{x\to\infty}\left(\frac{(x^2+5x+1)-(x^2-x+2)}{\sqrt{x^2+5x+1}+\sqrt{x^2-x+2}}\cdot\frac{\sqrt{x^2+5x+1}+\sqrt{x^2-x+2}}{\sqrt{x^2+5x+1}+\sqrt{x^2-x+2}}\right)=$$

$$=\lim_{x\to+\infty}\frac{x^2+5x+1-x^2+x-2}{x\left(\sqrt{1+\frac{5}{x}+\frac{1}{x^2}}+\sqrt{1-\frac{1}{x}+\frac{2}{x^2}}\right)}=$$

$$=\lim_{x\to\infty}\frac{x\left(6-\frac{1}{x}\right)}{x\left(\sqrt{1+\frac{5}{x}+\frac{1}{x^2}}+\sqrt{1+\frac{4}{x}+\frac{2}{x^2}}\right)}=$$

($\frac{1}{x}\to 0$, $\frac{1}{x^2}\to 0$, $\frac{4}{x}\to 0$, $\frac{2}{x^2}\to 0$)

$$=\lim_{x\to+\infty}\frac{6}{\sqrt{1}+\sqrt{1}}=\frac{6}{2}=3$$

N6 $\lim\limits_{x\to 0}\dfrac{\sin 3x}{\operatorname{tg} 5x}=\left[\dfrac{0}{0}\right]$

$\sin t\sim t$

$\operatorname{tg} t\sim t$

$$\lim_{t\to 0}\frac{3x}{5x}=\frac{3}{5}=0{,}6$$

N7 $\lim\limits_{x\to 0}\dfrac{1-\cos 4x}{\operatorname{arctg}^2(2x)}=\left[\dfrac{0}{0}\right]$;

$1-\cos t\sim\dfrac{t^2}{2}$ $\quad\operatorname{arctg} t\sim t$

$$\lim_{x\to 0} \frac{\frac{16x^2}{2}}{4x^2} = \lim_{x\to 0} \frac{8x^2}{4x^2} = \frac{8}{4} = 2$$

№8

$$\lim_{x\to 0} \frac{\cos 2x - \cos 5x}{x^2} = \left[\frac{0}{0}\right] =$$

$$\cos\alpha - \cos\beta = -2\sin\frac{\alpha-\beta}{2}\cdot \sin\frac{\alpha+\beta}{2}$$

$$= \lim_{x\to 0} \frac{-2\sin\left(-\frac{3x}{2}\right)\sin\left(\frac{7x}{2}\right)}{x^2} =$$

$$= \lim_{x\to 0} \frac{2\sin\frac{3x}{2}\sin\frac{7x}{2}}{x^2} =$$

$$= \lim_{x\to 0} \frac{2\cdot\frac{3x}{2}\cdot\frac{7x}{2}}{x^2} = \lim_{x\to 0} \frac{\frac{21}{2}x^2}{x^2} = \frac{21}{2} = 10{,}5$$

№9

$$\lim_{x\to 0} \frac{e^{4x}-1}{\ln(1+3x)} = \left[\frac{0}{0}\right]$$

$e^t - 1 \sim t$

$\ln(1+t) \sim t$

$$\lim_{x\to 0} \frac{4x}{3x} = \frac{4}{3}$$

№10

$$\lim_{x\to 1} \frac{\ln x}{\sqrt{x-1}} = \left[\frac{0}{0}\right]$$

$x-1 = t$

$x = t+1$

$(1+t)^k - 1 \sim kt$

$$\lim_{t\to 0}\frac{\ln(1+t)}{\sqrt{1+t}-1}=\lim_{t\to 0}\frac{t}{(1+t)^{\frac{1}{2}}-1}=\lim_{t\to 0}\frac{t}{\frac{t}{2}}=$$

$$=\lim_{t\to 0}\frac{2}{1}=2$$

№1

$$\lim_{t\to 0}\frac{\ln(1+t)}{\sqrt{1+t}-1}\left\{\lim_{t\to 0}\frac{t}{(1+t)^{\frac{1}{2}}-1}\right.$$

$$\lim_{x\to 0}\frac{\sqrt[3]{1+6x^2}-1}{1-\cos 2x}=\left[\frac{0}{0}\right]=$$

$$=\lim_{x\to 0}\frac{(1+6x^2)^{\frac{1}{3}}-1}{1-\cos 2x}=\lim_{x\to 0}\frac{\frac{2\cdot 6x^2}{3}}{\left(\frac{2x^2}{2}\right)}=$$

$$=\lim_{x\to 0}\frac{2x^2}{2x^2}=\frac{2}{2}=1$$

№2

$$\lim_{x\to 0}(1+5x)^{\frac{2}{x}}=[1^{\infty}]\quad\left|\lim_{t\to 0}(1+t)^{\frac{1}{t}}=e\right.$$

$$\lim_{x\to 0}\left(\left((1+5x)^{\frac{1}{5x}}\right)^{5x}\right)^{\frac{2}{x}}=e^{\lim_{x\to 0}\frac{10x}{x}}=e^{10}$$

№13

$$\lim_{x\to+\infty}\left(\frac{3x+2}{3x-4}\right)^x = [1^\infty]$$

$$\lim_{x\to 0}\left(1+\frac{1}{x}\right)^x = e$$

$$\lim_{x\to+\infty}\left(\frac{3x-4+6}{3x-4}\right)^x = \lim_{t\to+\infty}\left(\left(1+\frac{6}{3x-4}\right)^{\frac{3x-4}{6}}\right)^{\frac{6}{3x-4}\cdot x} =$$

$$= e^{\lim\limits_{x\to+\infty}\frac{6x}{2x-4}} = e^2$$

№14

$$\lim_{x\to\infty}(\cos 3x)^{\frac{1}{x^2}} = [1^\infty]$$

$$\lim_{x\to 0}(1+x)^{\frac{1}{x}} = e$$

$$(1-\cos t)\sim\frac{t^2}{2}$$

$$\lim_{x\to 0}(1-1+\cos 3x)^{\frac{1}{x^2}} =$$

$$= \lim_{x\to 0}\left(\left(1+(-(1-\cos 3x))\right)^{\frac{1}{-(1-\cos 3x)}}\right)^{-\frac{(1-\cos 3x)}{x^2}} =$$

$$= e^{\lim\limits_{x\to 0}\frac{-(1-\cos 3x)}{x^2}} = e^{\lim\limits_{x\to 0}\frac{-\frac{9x^2}{2}}{x^2}} = e^{-\frac{9}{2}} = e^{-4,5} =$$

$$= \frac{1}{e^{4,5}}$$